Muon Beam Production

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18/12/2025

Potential of J-PARC Muon Production

Acceleration of Positive Muons by a Radio-Frequency Cavity

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(Received 16 October 2024; accepted 21 April 2025; published 16 June 2025)

Acceleration of positive muons from thermal energy to 100 keV has been demonstrated. Thermal muons were generated by resonant multiphoton ionization of muonium atoms emitted from a sheet of laser-ablated aerogel. The thermal muons were first electrostatically accelerated to 5.7 keV, followed by further acceleration to 100 keV using a radio-frequency quadrupole with an intensity of 2×10^{-3} μ^+/pulse. The transverse normalized rms emittance of the accelerated muons in the horizontal and vertical planes were $0.85 \pm 0.25(\text{stat})^{+0.22}_{-0.13}(\text{syst})$ π mm mrad and $0.32 \pm 0.03(\text{stat})^{+0.05}_{-0.02}(\text{syst})$ π mm mrad, respectively. The measured emittance values demonstrated phase-space reduction by a factor of 2.0×10^2 (horizontal) and 4.1×10^2 (vertical) allowing good acceleration efficiency. These results pave the way to realize the first-ever muon accelerator for a variety of applications in particle physics, material science, and other fields.

DOI: 10.1103/PhysRevLett.134.245001

Aritome, S., Futatsukawa, K., Hara, H., Hayasaka, K., Ibaraki, Y., Ichikawa, T., ... & Yotsuzuka, M. (2025). Acceleration of positive muons by a radio-frequency cavity. *Physical Review Letters*, *134*(24), 245001.

rate of 25 Hz. In this experiment, the proton beam power at the Material and Life Science Experimental Facility was 940 kW, and the μ^+ intensity was 9.5×10^4 muons per pulse (μ^+/pulse). The profile of the surface muon beam was measured by a muon beam monitor with a gated image intensifier [16]. The horizontal and vertical rms beam sizes

10^5/pulse

cavity with a good acceleration efficiency.

Toward the application of this muon beam to various experiments, increasing its intensity is essential. The low muon rate in this experiment was due to the choice of a laser optimized for spectroscopy rather than ionization. By developing high-energy light sources operated at 122 and 355 nm dedicated to the ionization of muonium, the ionization efficiency is improved to around 10% [1]. The light source is currently under development at J-PARC [34]. In addition, a new muon beamline to deliver 70 times more intense surface muon beam is under construction [35]. In total, 10^5–10^6 increase of the muon rate is expected, potentially enabling the applications such as the muon $g-2$/EDM experiment at J-PARC [1].

$\times 10^5 (10^6)$

The results presented in this Letter show the phase-space reduction of a muon beam by more than 2 orders of magnitude. This enables broad applications of an accelerated positive muon beam from particle physics to various other scientific fields.

Acknowledgments—The authors would like to thank

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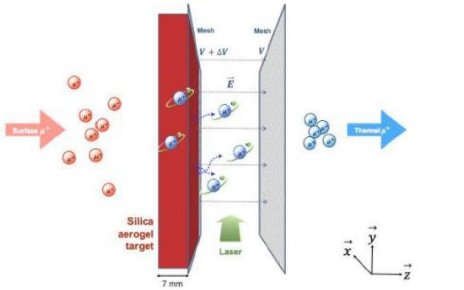
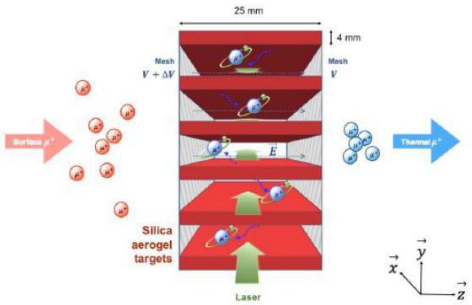
New ideas

Multi-layer target for Muonium production





Current design (single-layer)

- **Low Mu emission efficiency (0.0034):**
  - Muon stopping (0.418)
  - Vacuum emission (0.060)
  - Laser spatial constraint (0.269)

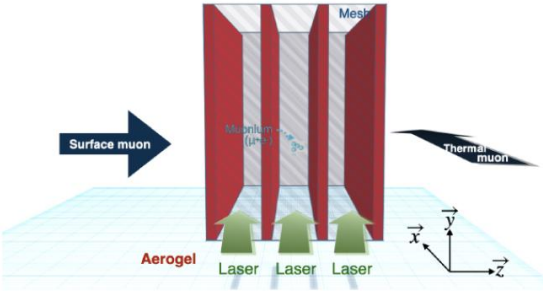
Novel multi-layer target design

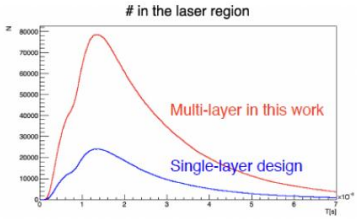
- Multi-layer targets stop incident muon
- Mu emits from upper and lower surfaces

New ideas

Multi-layer target for Muonium production

- Another version uses multi-layers facing the incident beam, resulting in a higher yield;
- The extraction is turned 90 degrees, making construction more challenging.





Simulation predicted even higher yields
(4 times if 4 layers → 8 times at max)

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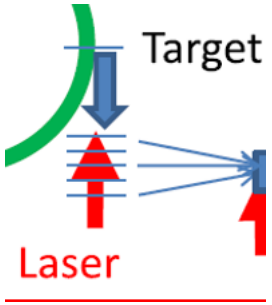
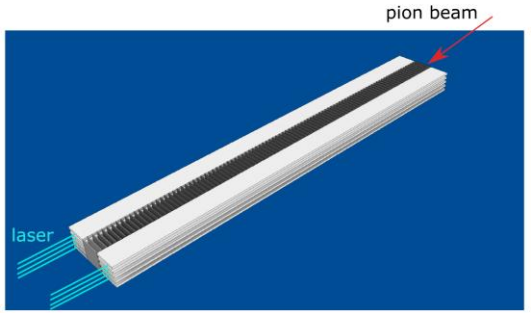
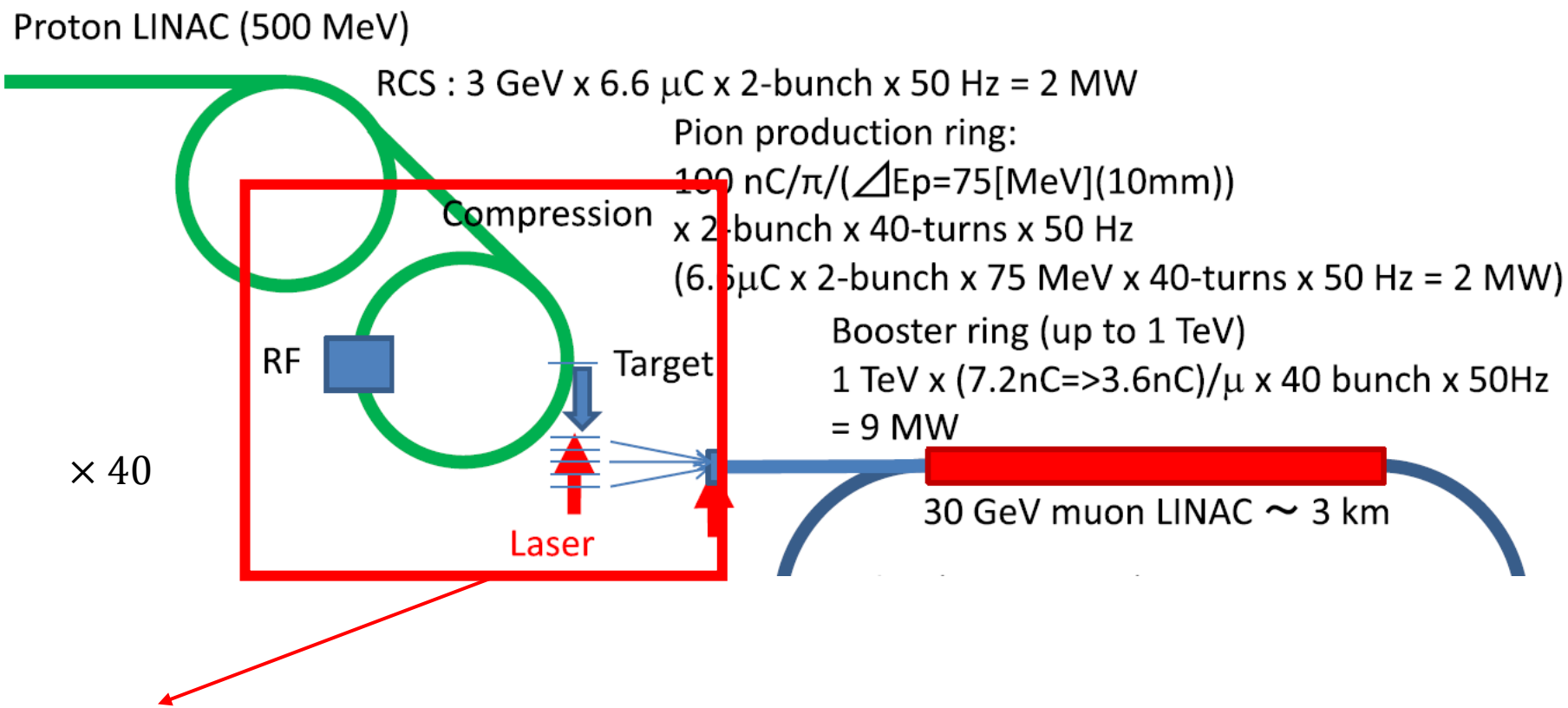
Courtesy of C. Zhang

Venanzoni, G. (2025, May 23). The g-2/EDM experiment @ J-PARC. HEP Annual Meeting, Liverpool, United Kingdom.





This multi-layer designs similar to the design muTRISTAN proposed



This pion production ring proposed by μTRISTAN

•The proton beam circulates for 40 turns in the pion production ring, bunches striking the target 40 times and 40 muon bunches are generated

Hamada, Y., Kitano, R., Matsudo, R., Takaura, H., & Yoshida, M. (2022). μTRISTAN. *Progress of Theoretical and Experimental Physics*, *2022*(5), 053B02. https://doi.org/10.1093/ptep/ptac059

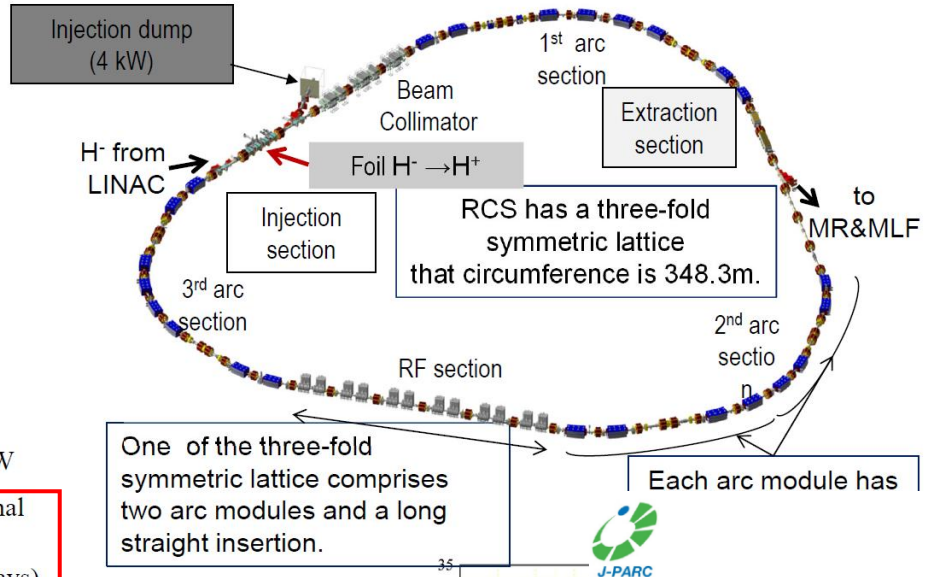
3GeV-RCS in J-PARC

Design parameters

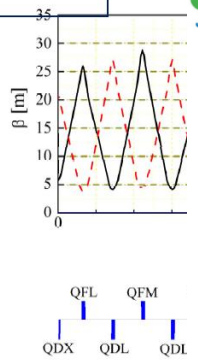
Circumference	348.333 m
Superperiodicity	3
Harmonic number	2
F_{rev}	0.61-0.84 MHz
F_{rf}	1.23-1.67 MHz
Injection energy	400 MeV
Extraction energy	3 GeV
Repetition rate	25 Hz
Particles per pulse	8.3e13 with 1 MW
Output beam power	0.75 MW in ordinal operation (1 MW for two days)
Transition gamma	9.14
Number of dipoles	24
quadrupoles	60 (7 families)
sextupoles	18 (3 families)
steerings	52
RF cavities	12

$\times 50$

50 Hz at muTRISTAN (2 MW)



Note:
Beam power is limited not the beam loss but the durability of the neutron target (and cooling water performance in summer).



$10^{13} - 10^{14}\ \mu^+/s$ could be achieved!!

Yamamoto, K. (2021, May 20). High power study of J-PARC Rapid Cycling Synchrotron. 1st Muon Community Meeting, J-PARC Center Accelerator Division.

Summary

Operation

- RCS is almost continuing stable user operation.
  - At present, ***6.2e13 ppp (750-kW)*** beam delivers to the MLF and ***6.7e13 ppp (515-kW in MR)*** beam to the MR.
  - So far, the neutron target limits the operation beam power. We will increase the beam power with carefully monitoring the target status and the beam loss.

High power study

- We tried 1-MW, 2-day continuous operation, but it revealed some issues to achieve stable user operation with 1-MW.
- From our experiences, design of the injection is quite important to realize multi-MW beam.
- Trials beyond 1-MW were carried out. Results indicated RCS has enough capability to accelerate 1.5-MW beam if RF system would be reinforced.